

- A constraint for an element MAY use the name of an attribute in the attribute table, when annotated with a reference to the element.
- A constraint for an element for command field of response commands in a cluster MAY use the name of a command field in the associated request command, when annotated with a reference to the element.

7.18.3.5. List and String Constraints

The constraints for Lists and Strings SHALL indicate the capability of the server with respect to the number of entries of the associated List or size of the associated String.

The minimum number of entries for a list or the size of a string SHALL be 0 (zero), unless redefined using the above notation.

A **range** constraint for a List or String defines the number of entries or size, inclusive, for the element that SHALL be supported by the server. Clients writing to or otherwise affecting changes in elements with a range constraint MAY supply a list of any length or string of any size provided that the length of the list or size of the string are between *x* and *y* inclusive.

A **limit** constraint for a List or String defines a single-ended limit to the number of entries or size, inclusive, for the element that SHALL be supported by the server.

Notes on clients writing to or affecting changes in elements with a limit constraint:

- *min*: Clients writing to or otherwise affecting changes in elements with this constraint MAY supply fewer elements than the constraint value. For example, a client writing to the a list attribute with a constraint of *min 4* MAY write a list with the number of items being between 0 (zero) and 4 inclusive, or more than 4 provided that (the client has knowledge that) the server supports more than 4 items.
- *max*: Clients writing to or otherwise affecting changes in elements with this constraint MAY supply fewer elements than the constraint value. For example, a client writing to a string attribute with a constraint of *max 16* MAY write a string of any size between 0 (zero) and 16.

A **union** constraint for a List or String is a comma-delimited set of constraints. A union constraint SHALL only have one minimum (*min x*) constraint and one maximum (*max y*) constraint. A union constraint SHALL NOT define a range below the minimum constraint or range greater than the maximum constraint, including the defined **minimum** (*min*) and **maximum** (*max*) for the data type.

A constraint on a list or string data means that the data SHALL always be indicated within that constraint. A constraint on a **writeable** list or string data means that the data SHALL support writing within the constraint, and SHALL NOT support writing outside the constraint.

The actual number of list entries or string size for an element MAY differ from the capability as determined by the constraints but SHALL NOT violate the constraints for the element.

For example, a string attribute may be defined with a constraint of **5 to 7**:

- A value of "bear" would violate the constraints as the size of 4 is less than the lower end of the constraint range, which is 5.